

Resolving organoid brain region identities by mapping single-cell genomic data to reference atlases

Fleck et al., Cell Stem Cell(2021)

Journal Meeting

2024. 11. 17

변가은

KOREA
UNIVERSITY



Prof. Dr. Barbara Treutlein

barbara.treutlein@bsse.ethz.ch



Quantitative
Developmental
Biology Lab

**QuaDBio Lab**
Quantitative Developmental Biology Lab
61 followers Switzerland <https://bsse.ethz.ch/qdb> @TreutleinLab

Overview Repositories 22 Projects Packages People 1

Popular repositories

scRNAseq_analysis_vignette Public
Tutorial for scRNA-seq data analysis beginners using R
154 58

Pando Public
Multiome GRN inference.
R 110 21

scMultiome_analysis_vignette Public
Tutorial of single-cell RNA-ATAC multiomic sequencing data analysis
47 13

VoxHunt Public
VoxHunt: Resolving human brain organoid heterogeneity through single-cell genomic comparison to spatial brain maps
HTML 41 5

primate_cerebral_organoids Public
Reusable scripts and functions archive for the paper "Organoid single-cell genomic atlas uncovers human-specific features of brain development" (<https://www.nature.com/articles/s41586-019-1654-9>)
R 33 3

simspec Public
Calculation of Reference/Cluster Similarity Spectrum (RSS/CSS)
R 30 3



Overview

1. Introductions

- What are Brain Organoids and current challenges in brain organoid research
- Brief Description of Multi-modal transcriptomics data

2. Voxhunt: A computational toolkit for mapping multi-modal organoid data

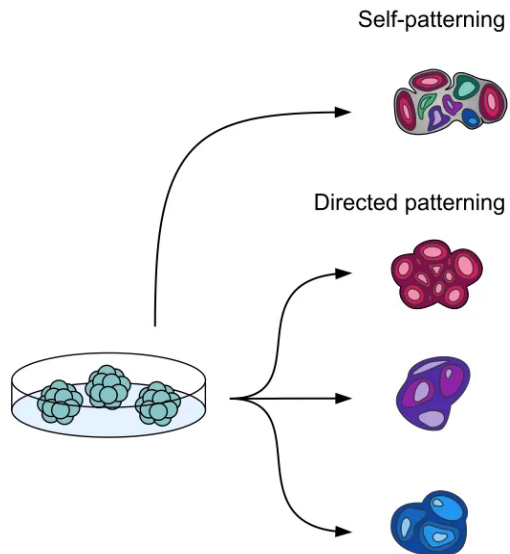
- How it works? – Methods and Key results
- Application and Insights

3. Discussion

Brain Organoids

What are Brain organoids and current challenges in brain organoid research

- **Brain Organoids**
 - **it is not yet possible to create a stereotypic organoid** containing multiple brain structures to study neuronal migration or enable formation of inter-region neuronal connections



- **Self- patterning strategy**

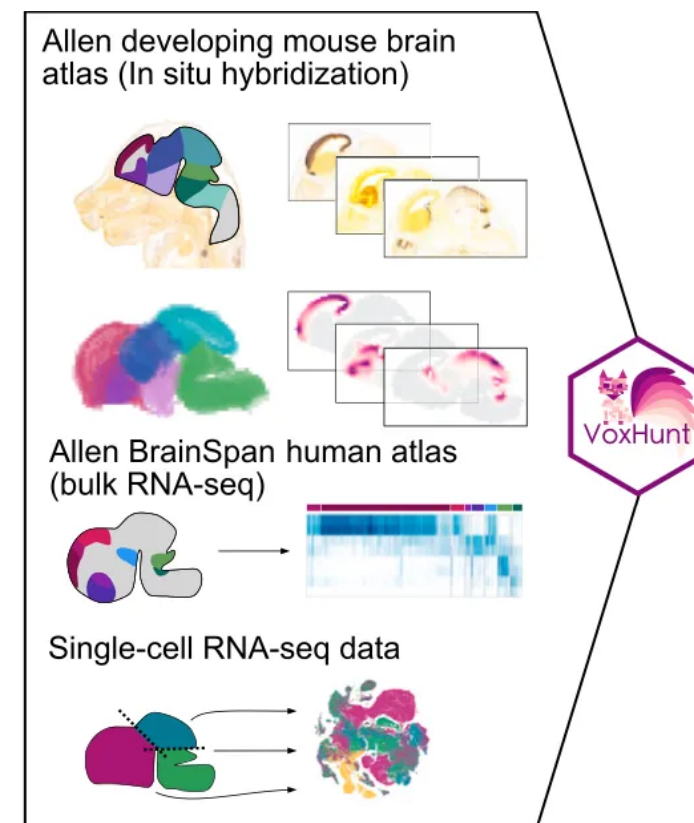
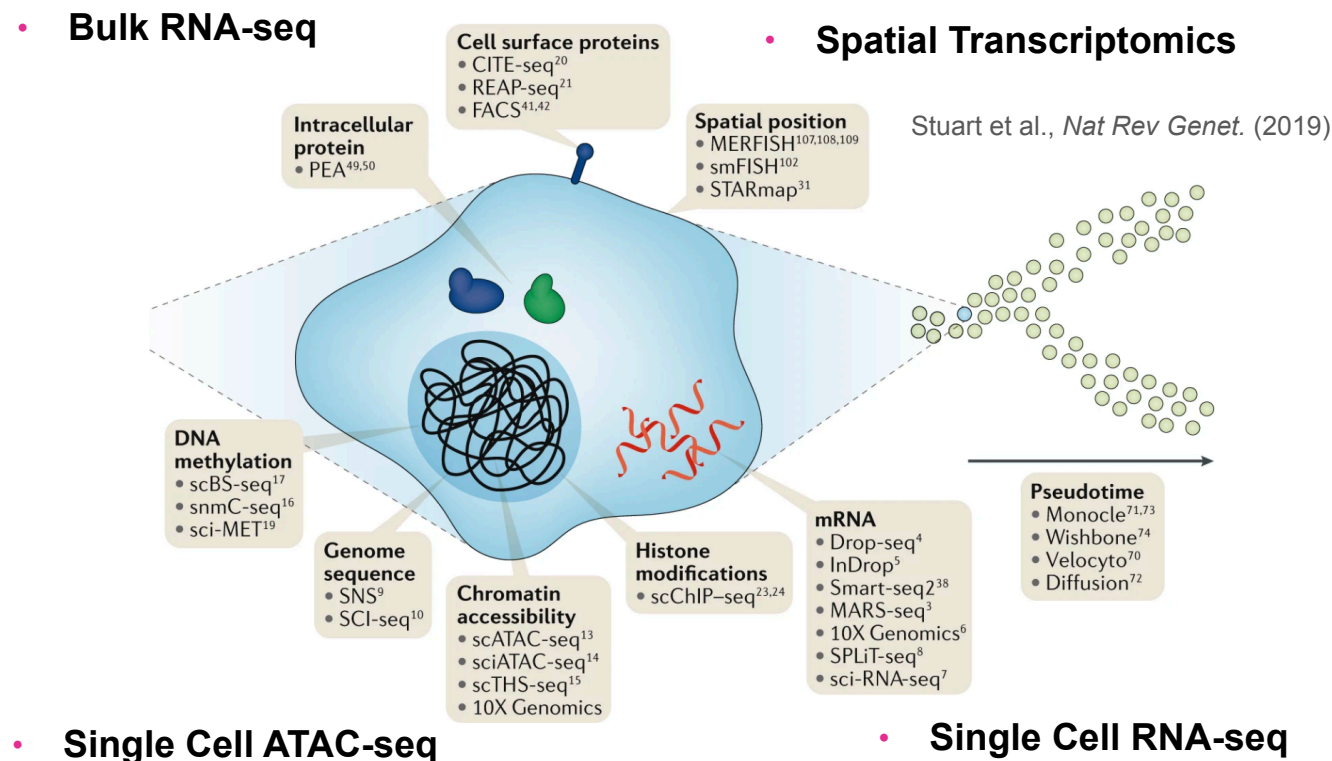
: Guiding PSCs to form a layer of neuroepithelial stem cells which can **differentiate** into neurons

- **Direct strategy**

: Using Signaling molecules or directive cues to **control** patterning of the neuroepithelium

Interpreting gene expression and regulatory patterns in organoids still remains a challenging area

Multi-modal Transcriptomics Data for integrative analysis of cellular systems



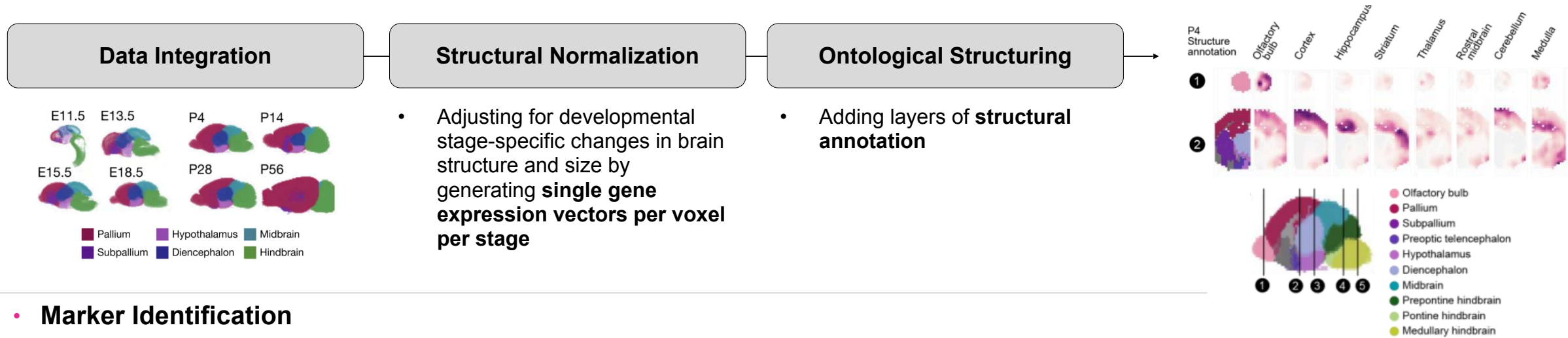
- We can assess brain organoid **patterning**, **developmental state**, and **cell identity** comparing with annotated reference data

Results

Spatiotemporal Voxel Mapping Framework

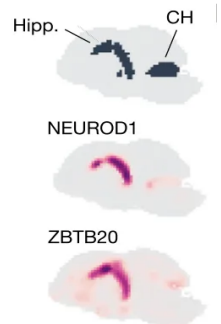
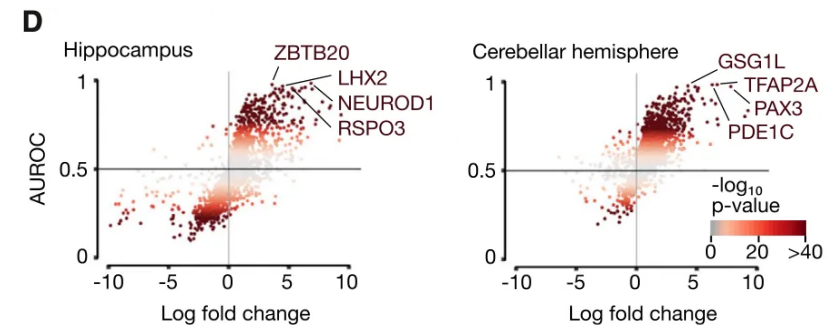
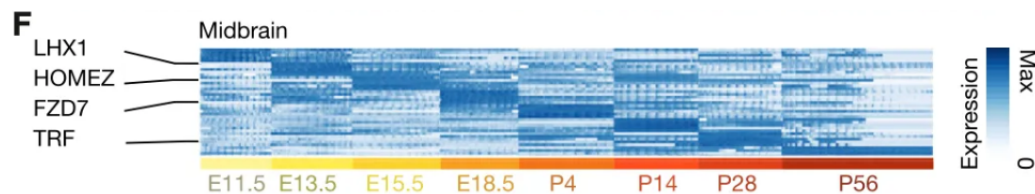
Reference

- in situ hybridization data of the mouse brain in **8 different developmental stages** from Allen Developing Mouse Brain Atlas



Marker Identification

- DE analysis to identify **brain structure-specific marker genes** using the **single-feature AUROC** metrics



Results

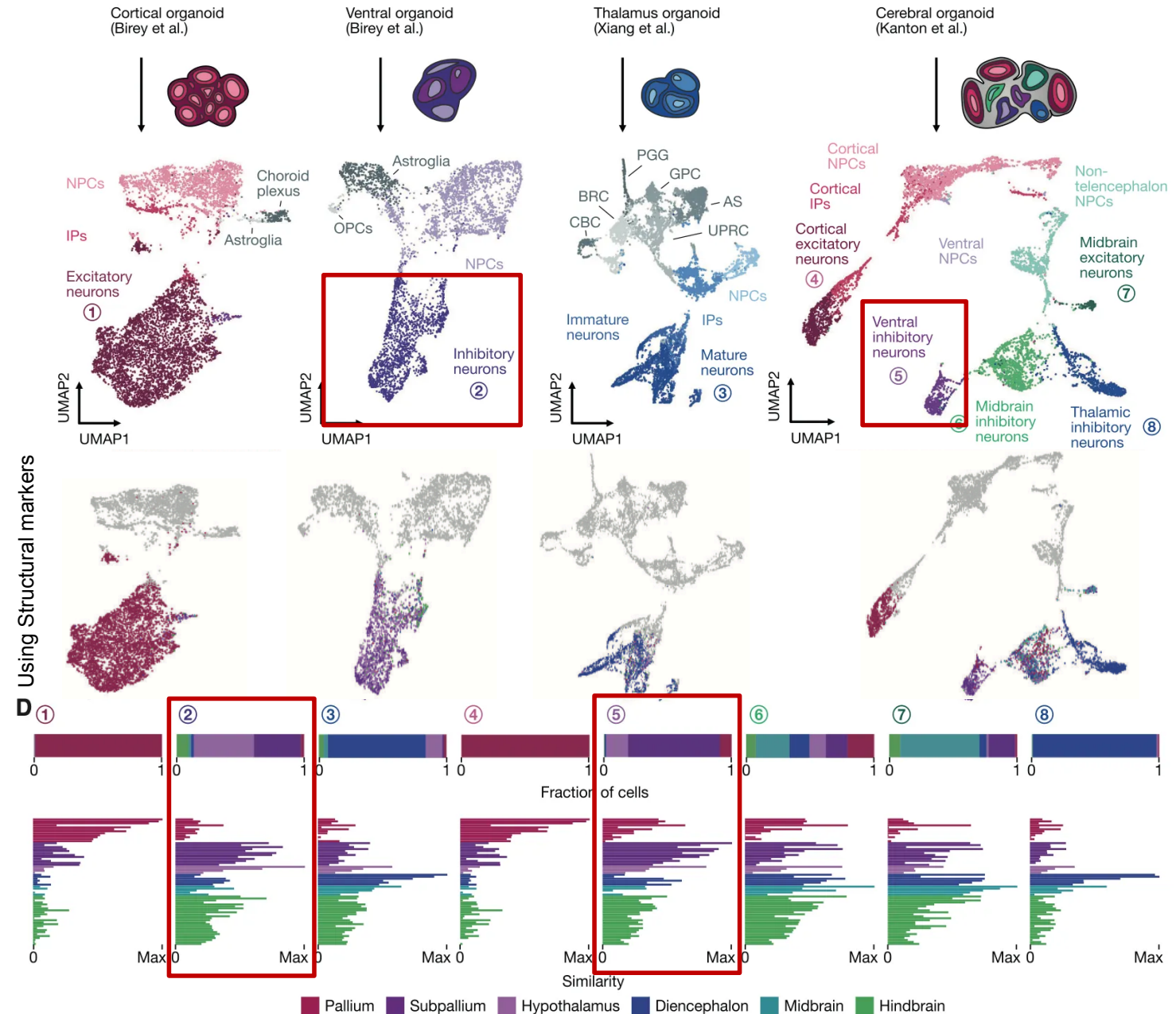
Organoid Region Composition Annotation

1. Mapping(Unsupervised annotation) with 4 different organoid **scRNA-seq** data

- Thalamus, Cortex, Ventral telencephalon, Cerebral(unpatterned)

2. **Patterning** structure-specific organoid's neuronal cells with canonical structural marker genes

3. Comparing Neuronal clusters with mammalian brain's ISH data

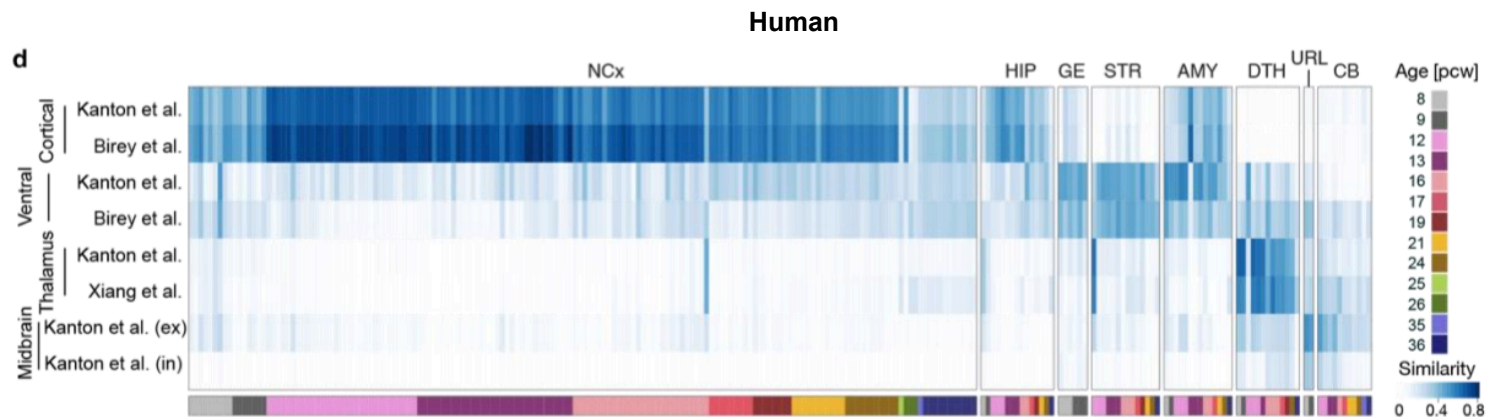


Results

Organoid Region Composition Annotation

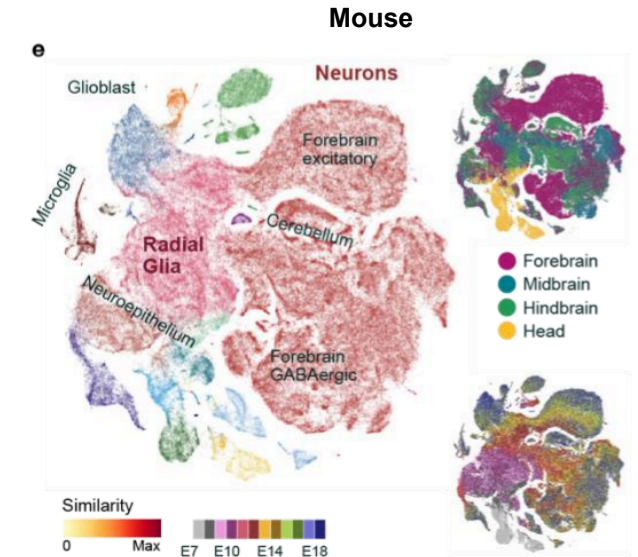
With Human reference data (Brainspan) & Mouse brain

3. comparing organoid-derived neurons to time course **bulk RNA-seq data** from the **human brain**(Thompson et al., 2014) and **single cell atlas** of the **developing mouse brain**(La Manno et al., 2020)



NCx: Neocortex, HIP: Hippocampus, GE: Ganglionic eminence, STR: Striatum, AMY: Amygdala, DTH: Dorsal thalamus, URL: Upper rhombic lip, CB: Cerebellum

- Lack of midbrain sample



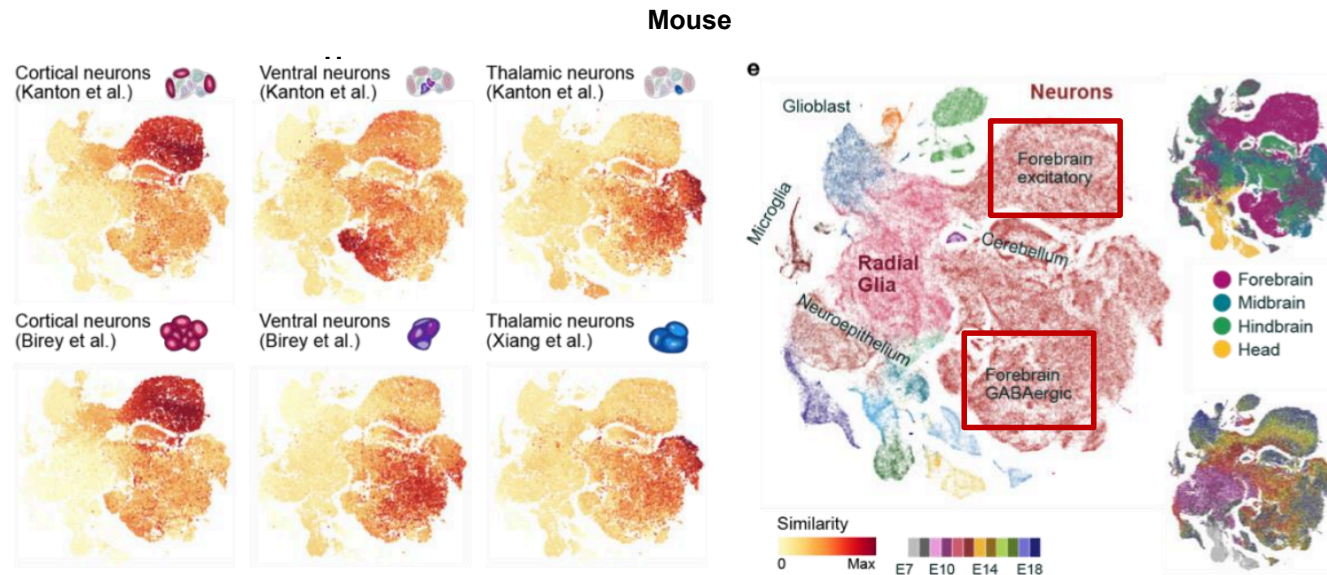
- Providing annotation not covered in ISH data

Results

Organoid Region Composition Annotation

With Human reference data (Brainspan) & Mouse brain

3. comparing organoid-derived neurons to time course **bulk RNA-seq data** from the **human brain** (Thompson et al., 2014) and **single cell atlas** of the **developing mouse brain** (La Manno et al., 2020)



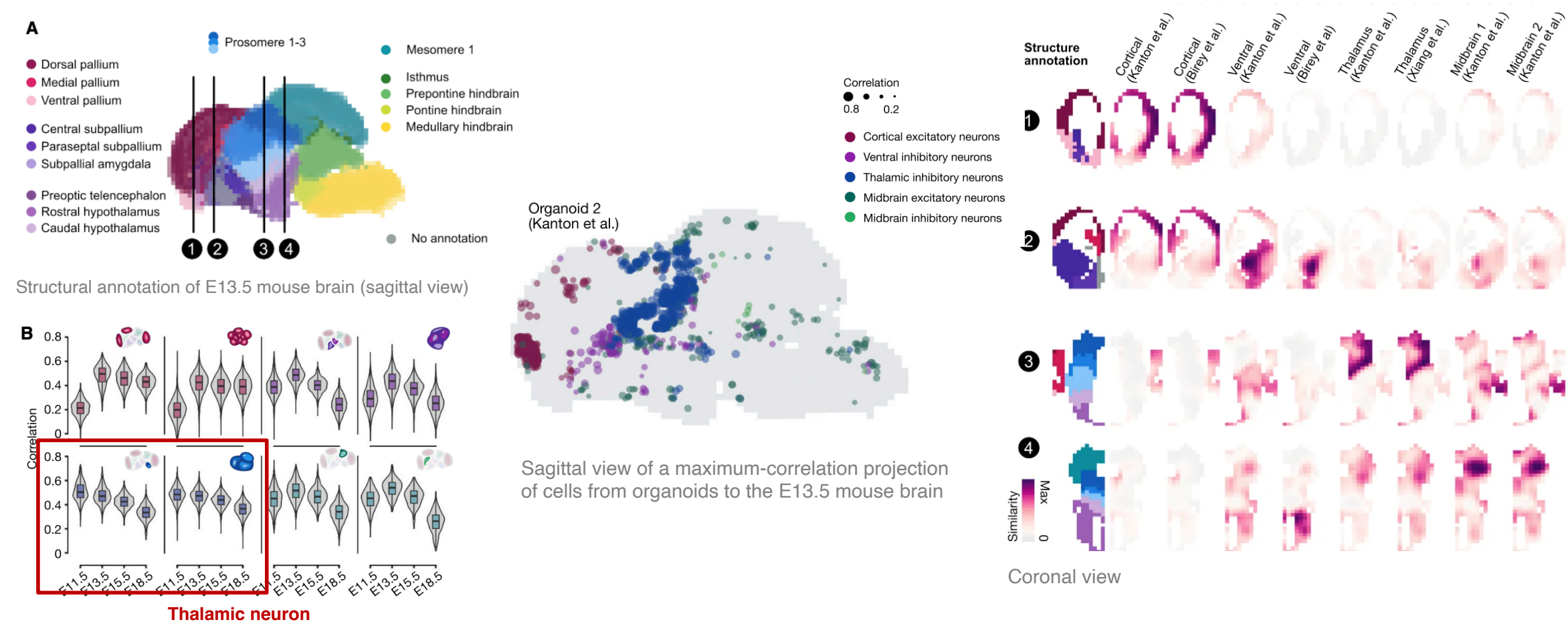
- Providing annotation not covered in ISH data

Comprehensive whole-brain scRNA-seq datasets from human and mouse will be valuable references for functionally annotating neuronal and non-neuronal cell types.

Results

Spatial projection of organoid cell transcriptomes

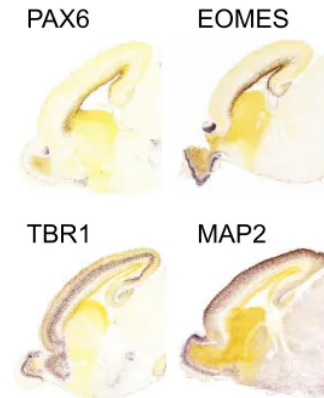
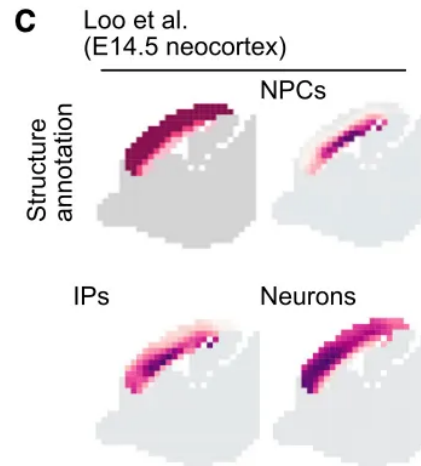
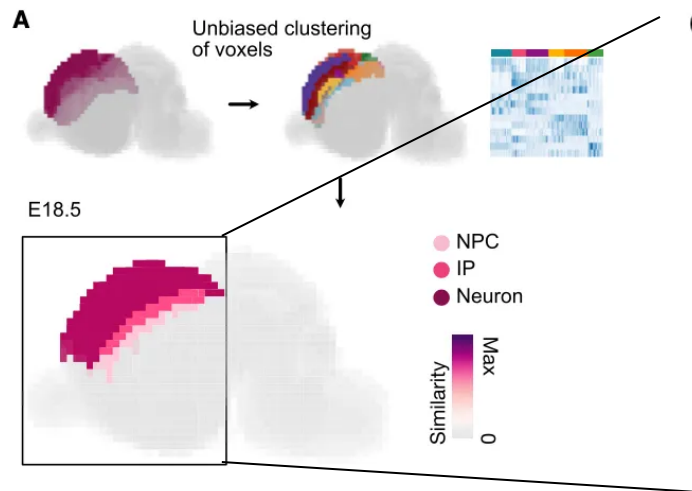
- To assess the **developmental stage and spatial localization** of organoid-derived cells
- **Correlation analysis** of organoid clusters to fetal developmental stages (E11.5–E18.5) with 3D reference dataset



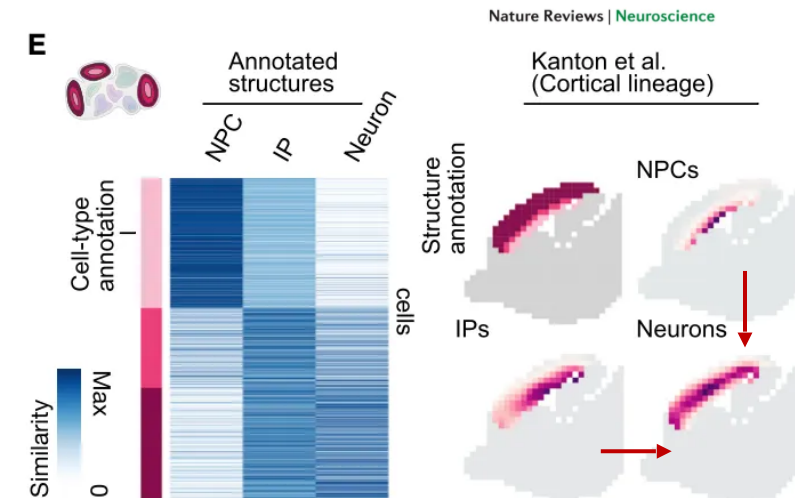
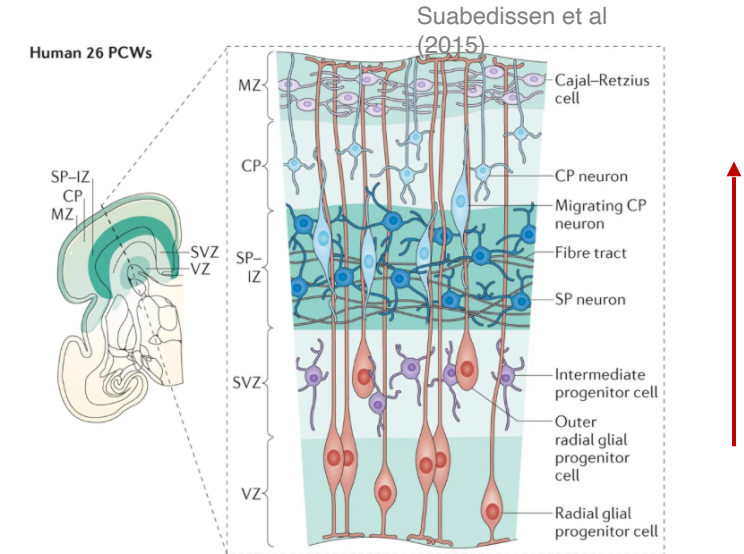
Results

Annotating organoid cell identity at substructure resolution

- Unbiased clustering of voxels in **Dorsal pallium**
- Annotating with **canonical marker gene expression**



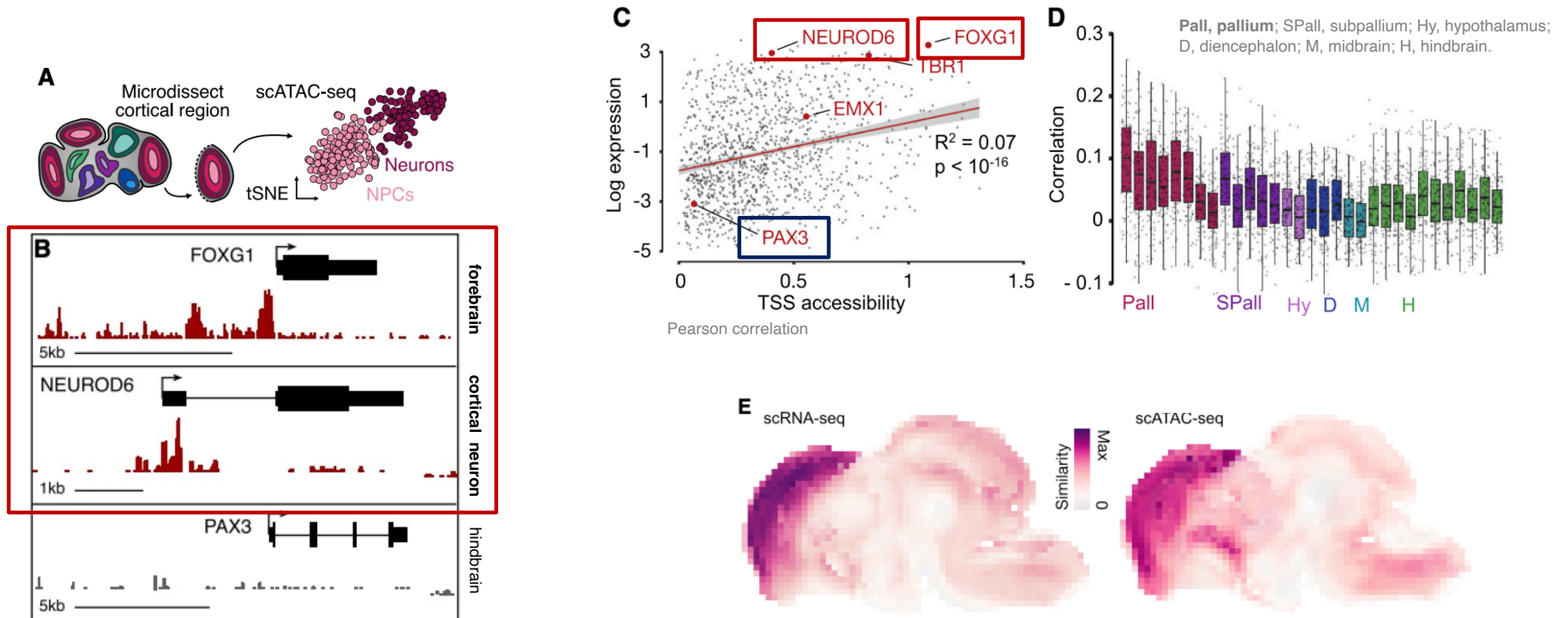
ISH images of marker gene expression



Results

Chromatin Accessibility Analysis

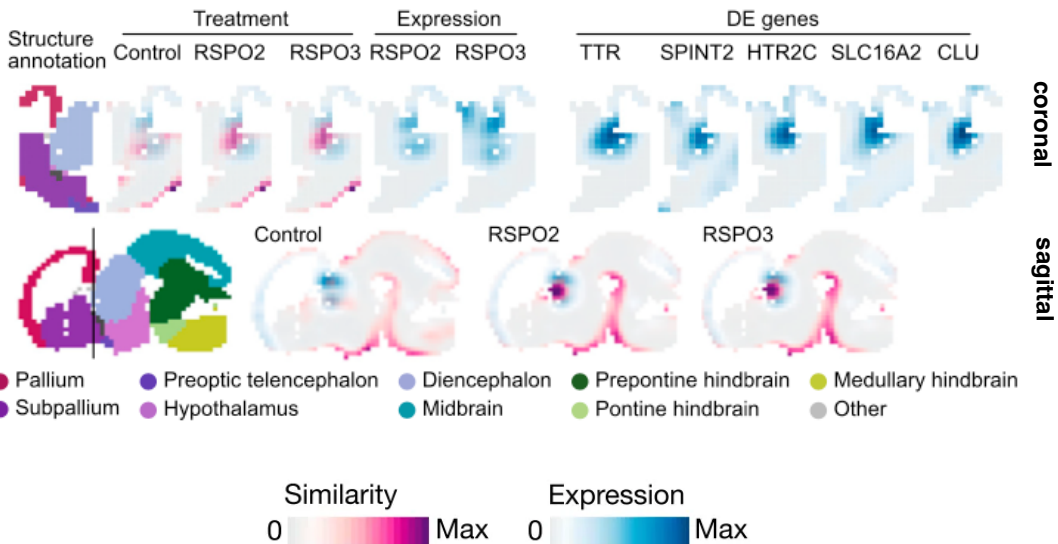
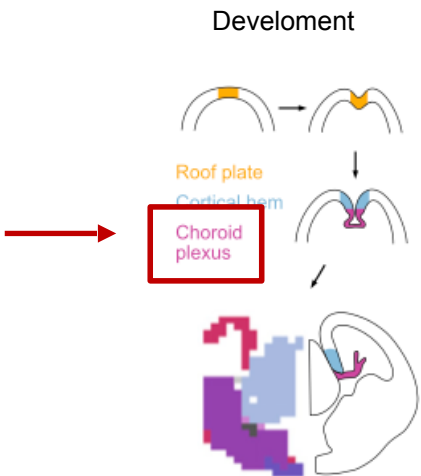
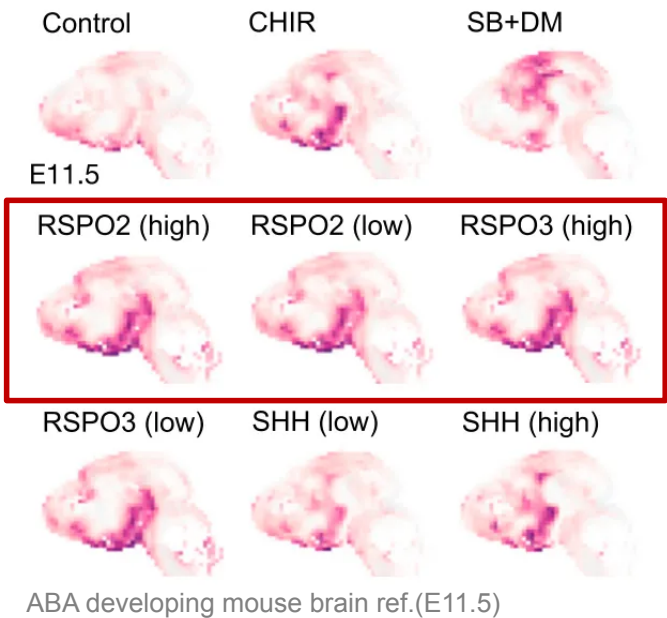
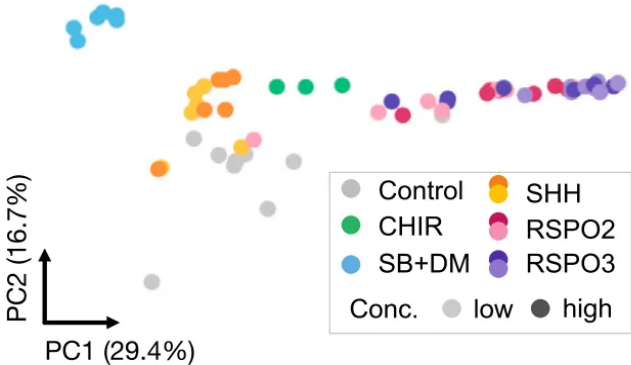
- Mapping with scATAC-seq data with the pair of scRNA-seq & scATAC-seq data from 2 to 4M cerebral organoids(Kanton et al., 2019)
- Calculating Accessibility score in TSS



Results

Application :: Manipulating Organoid Patterning

- Can assess the culture condition manipulations output
 - R-spondin 2 (RSPO2), R-spondin 3 (RSPO3),
 - sonic hedgehog (SHH)
 - an activator of the Wnt pathway (CHIR99021)
 - two small molecules for dual SMAD inhibition (SB431542 and dorsomorphin)
- With Bulk RNA-seq data



A potential hint into how local morphogen expression impact the development within neighboring structures

Discussion

Key takeaways

- ISH data 및 Multi-modal transcriptomics data를 활용한 Integrative analysis
- Multi-modal data를 활용한 Organoid assessment와 developmental analysis 적용

Limitation & Scalability

- Whole brain에 대한 single cell 수준의 spatial reference가 부족
- Marker gene expression에 기반한 Annotation

Thank You